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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 731 — Past Questions & Materials
by Renance

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Paper 1 of 1 — CONTROL NOTE EEE731 and EE411

Level 400 • First Semester • Handout

EEE 731: CONTROL SYSTEMS

1. INTRODUCTION

Control systems are an integral part of modern society. Numerous applications are all around us: The rockets fire, and the space shuttle lifts off to earth orbit; in splashing cooling water, a metallic part is automatically machined; a self-guided vehicle delivering material to workstations in an aerospace assembly plant glides along the floor seeking its destination. These are just a few examples of the automatically controlled systems that we can create. refers to measuring the value of the controlled variable of the system and applying the manipulated variable to the system to correct or limit deviation of the measured value from a desired value.

We are not the only creators of automatically controlled systems; these systems also exist in nature. Within our own bodies are numerous control systems, such as the pancreas, which regulates our blood sugar. In time of “fight or flight,” our adrenaline increases along with our heart rate, causing more oxygen to be delivered to our cells. Our eyes follow a moving object to keep it in view; our hands grasp the object and place it precisely at a predetermined location

Basic terms

Controlled variable and manipulated variable: Controlled variable refer to the quantity or condition being controlled or measurement and this is typically the output of the system. While a manipulated variable is the quantity controlled by the controller in order to affect the controlled parameter’s value.

Plants: Simply a piece of equipment, and more concisely a set of machine parts assembled to perform a given function together. Thus, any physical object (such as chemical reactors, spacecraft heating furnace etc.) being controlled can be referred to as a plant.

Systems: A system is an assemblage of components that synergies to perform a specific objective. The system is not limited to physical ones and may be abstract in form or dynamic phenomena.

Disturbances: This refers to an adverse input signal that affects the output of a system. Disturbances generated may be exogenous (outside the system) or endogenous (inside the system).

1.1 Open Loop Control systems

Control system in its simplest form comprises of subsystems and processes (or plants) assembled with the aim of obtaining a desired output with a desired performance. Figure 1 shows the open loop control system which is the simplest form of control system.

Figure 1. Open loop Control

The control system exists as an integral part of the current modern society and they are widespread. They can be seen for example in an Elevator system where the speed of the

elevator and the floor leveling accuracy is design to ensure passengers comfort.

More so, in the elevator system, the push button is an input which denotes the desired output shown as a step function in Figure 1.2. The performance of the elevator can be seen from the elevator response curve in the figure.

Figure 1.2. The elevator response curve

Two major evaluation measures of performance are apparent: (1) the transient response and
(2) the steady-state error.

In the above example, passenger comfort and passenger patience are dependent upon the transient response.

If this response is too fast, passenger comfort is sacrificed; if too slow, passenger patience is sacrificed. The steady-state error is another important performance specification since passenger safety and convenience would be sacrificed if the elevator did not level properly.

Advantages of open loop control systems

1. Simple construction and ease of maintenance.
2. Cheaper than closed loop systems
3. No problem of stability
4. More convenient when the output performance is difficult to estimate (e.g. Washing machine measurement of cloth cleanliness will be expensive).

Disadvantages of Open loop Control Systems

1. Prone to calibration error due to the impact of disturbances
2. The necessity for repeated recalibration.

1.1.1 Benefits of Control Systems

1. Power amplification
2. Remote Control
3. Convenience of input form
4. Compensation for disturbances

1. Power Amplification

For example, a radar antenna positioning requires power amplification (power gain) of the input signal for the successful rotation of a knob or gear system which translates to the actual output rotation.

2. Remote Control

Robots are very useful in remote or dangerous environments. For example, robot-controlled arm can be used for decontamination such as picking up a radioactive material.

3. Convenience of input form

Systems can provide convenience by changing the form of the input. For example, in a temperature control system, the input is a position on a thermostat. The output is heat.

Thus, a convenient position input yields a desired thermal output.

4. Disturbance compensation

The control system can detect disturbances and compensate for such disturbance for example if the wind forces the antenna to veer from its desired angle the system repositions the antenna commandeered by the input.

2.1.2 Examples of Control Systems

1. Computer numeric controlled machining (Lathe) tools in the manufacturing industry
2. Cruise Control in Automobiles
3. Auto pilot in airplanes
4. Assembly robots in the manufacturing industry

2.1.3 Feedback Control System

Feedback Control refers to an operation which reduces the difference (deviation) between the output of a system and the reference input based on the difference in the presence of disturbances. E.g. the temperature in a room can be measured and compared to a reference (desired) temperature and a thermostat is turned ON/OFF depending on the room temperature and regardless of the outside temperature to ensure comfort.

Closed Loop Control System

Example of a Closed Loop system

The watts speed governor for engine is shown in Figure 1.3,

Figure 1.3: The watts speed governor for engine

The difference between the desired speed and the engine speed is used to adjust the amount of fuel admitted into the engine.

In Figure 1.4, an automatic controller compares the liquid level in the tank with the desired level and adjusts the flow rate via the pneumatic valve if the level is insufficient or an error exists.

Figure 1.4 (a) Liquid Level Control system (b) Block diagram

Advantages of Closed loop systems

1. More accurate in the presence of non-linearities
2. Sensitivity may be minimised and system stability enhanced
3. Less prone to noise or disturbances

Disadvantages of Closed loop system

1. Costly compared to open loop control systems
2. Feedback may lead to oscillatory response or an unstable system
3. More care is required in the design of closed loop systems

Analysis and design objectives

There are three major objectives in analysis and design of control systems these are :
produce the desired transient response, reduce steady state error and ensure stability.

Analysis is the process by which a system's performance is determined. For example, we evaluate its transient response and steady-state error to determine if they meet the desired specifications.

Design is the process by which a system's performance is created or changed. For example, if a system's transient response and steady-state error are analyzed and found not to meet the specifications, then we change parameters or add additional components to meet the specifications.

Transient response: This is the response of the system which exhibits dynamics with a rate of change in time. Thus, as in the case of the elevator example, a slow transient response translates to longer waiting time in the elevator which may result in impatience on the user's path. While a fast-transient response can cause harm or damages to both the elevator and the user. Therefore, the transient response must stay within the acceptable or desired specification to guarantee both comfort and safety.

Steady state: This response is usually what remains after the transient dynamics decays and the systems response stays constant for evermore. In the elevator example the steady state response will be the final floor where the elevator stopped.

References